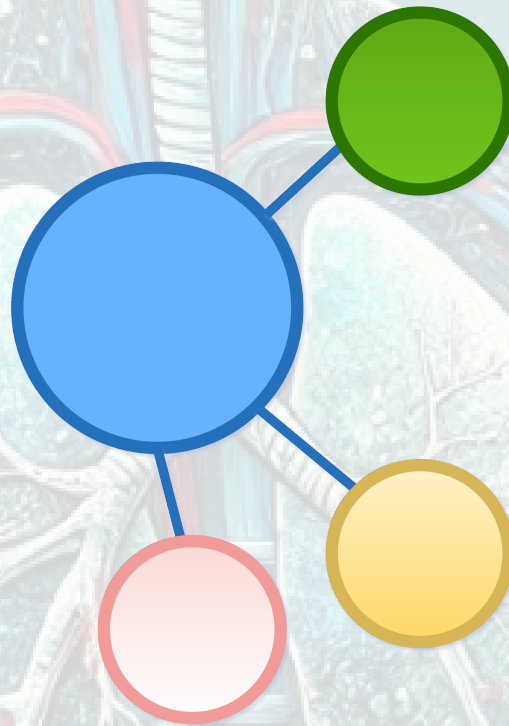


Installation Guide for ORTHOGATHER



OrthoGather

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1 Introduction

This document provides a detailed, step-by-step guide for installing and configuring **OrthoGather** on a local machine.

OrthoGather is a locally hosted web application that streamlines comparative proteome and interspecies functional analysis by integrating orthogroup inference, interactive visualisation, and gene ontology enrichment into a unified platform. It enables users to download proteomes directly from the UniProt database, run *OrthoFinder* locally, and perform both comparative and functional analyses through an intuitive browser-based interface built with *Flask*.

The platform supports three main input modes — *External Data Upload*, *New Analysis*, and *Preselected Dataset* — allowing users to either import existing *OrthoFinder* results, generate new analyses from UniProt proteomes, or explore the included example dataset. Outputs include UpSet plots summarizing orthogroup distribution, Excel exports with UniProt identifiers, and GO enrichment visualizations.

OrthoGather has been developed in Python 3.7 and is designed to run locally on macOS and Linux systems compatible with *OrthoFinder*. It can also operate under Windows using a virtual environment or the Windows Subsystem for Linux (WSL). The platform requires only an internet connection for downloading UniProt proteomes or Gene Ontology Annotation (GOA) files; once the data are retrieved, the analyses can be performed fully offline.

Note: If you are reinstalling the tool due to an update, there are two approaches:

1. Remove the previous installation folder and perform a clean setup from scratch (this does not require reinstalling *Python* or *Git*).
2. Follow Section 7 to update the existing installation and verify that all components have been correctly revised.

Option 2 can always be used for updates; however, if any issues arise, reverting to complete reinstallation (Option 1) is recommended.

This document serves as an **installation guide** and does not replace the user manual or the methodological description available in the main OrthoGather publication.

For questions or technical issues during the installation, please contact:
carlos.vivasrodriguez@ucdconnect.ie.

2 System Requirements and Installation Overview

Before starting the installation, please review this section carefully. It provides an overview of the software requirements, installation methods, and tools used to configure **OrthoGather** on your system.

Supported Operating Systems

- **macOS** (tested on macOS Ventura and Monterey, Intel and Apple Silicon via Rosetta)
- **Linux** (tested on Ubuntu 22.04 and Debian-based distributions)

- **Windows 10/11 via WSL (Windows Subsystem for Linux)** — recommended over native Windows

Why Conda and Micromamba are Required

OrthoGather is built on Python 3.7 and integrates multiple bioinformatics tools such as *OrthoFinder*, *GOATOOLS*, and *Flask*. To ensure consistent versions across systems, the tool relies on isolated environments created through either:

- **Conda** (used on macOS): a Python-based environment manager, ideal for Apple systems and required for Rosetta emulation on Apple Silicon.
- **Micromamba** (used on Linux/WSL): a C++ reimplement of Conda that performs the same tasks but is far faster and lighter, solving dependency issues in seconds instead of minutes.

Both systems use the same package channels (*conda-forge*, *bioconda*) and produce the same environment — named **orthogather37**.

In summary: macOS installations rely on Conda (Intel mode through Rosetta), while Linux and WSL installations use Micromamba for speed and stability.

Available Installation Methods

There are **three alternative ways to install OrthoGather**, depending on user experience and system type:

1. **Automated installation for Linux/WSL:** Using the script `install_orthogather_wsl.sh`, included in the repository. This is the recommended option for most users on Windows (via WSL) or Linux.
2. **Automated installation for macOS:** Using the script `install_orthogather_mac.sh`, also included in the repository. This script handles Intel/Rosetta compatibility automatically.
3. **Manual installation (for advanced or bioinformatics users):** A `requirements.txt` file is provided for those who prefer to install dependencies manually using their own environment or cluster configuration. This option assumes familiarity with Conda or virtual environments:

```
conda create -n orthogather37 python=3.7
conda activate orthogather37
pip install -r requirements.txt
```

Manual installation is flexible but not officially supported for troubleshooting.

Git and Git LFS: Essential Components

Git and **Git LFS (Large File Storage)** must be installed before cloning the repository.

- **Git** retrieves the OrthoGather source code and scripts from GitHub.
- **Git LFS** ensures large reference files (e.g. `proteomes_list.json`, ~170 MB) are downloaded correctly.

To install them:

- On Linux/WSL:

```
sudo apt update
sudo apt install -y git git-lfs
git lfs install
```

- On macOS (with Homebrew):

```
brew install git git-lfs
git lfs install
```

Verify installation:

```
git --version
git lfs version
```

Next Steps

Once Git and Git LFS are installed, proceed to the next section to clone the repository and access the installation scripts.

3 Downloading the GitHub Repository

Once Conda, Git, and Git LFS are installed, you can download the full OrthoGather source code and data.

3.1 Clone the repository

Choose a permanent location (for example, your home directory or a “Projects” folder), then run:

```
cd ~
git clone https://github.com/CarlosVivasR/OrthoGather.git
cd OrthoGather
```

This creates a local `OrthoGather` directory containing the code, HTML templates, launch scripts, and configuration files such as `environment.yml`.

3.2 Download large reference data (Git LFS)

OrthoGather includes a large proteome metadata file (`static/Proteomes_json/proteomes_list.json`, typically 160–180 MB). This file is stored via Git LFS and will *not* download correctly unless Git LFS is initialised.

Run:

```
git lfs pull
```

Confirm that the file was downloaded correctly (it should be large, not just a tiny 3-line placeholder):

```
ls -lh static/Proteomes_json/proteomes_list.json
```

4 Installation on macOS

This section describes how to install and configure **OrthoGather** on macOS systems, covering both Intel processors and Apple Silicon (M1/M2/M3) through Rosetta emulation. All steps are automated through the included installer script `install_orthogather_mac.sh`.

Why a special setup is needed on macOS

Python 3.7 and *OrthoFinder* are distributed as **Intel (x86_64)** binaries. Apple Silicon Macs use the ARM architecture, which is incompatible with these binaries. To solve this, OrthoGather runs under **Rosetta 2**, an Apple emulation layer that allows Intel software to run seamlessly on ARM chips.

Installing Conda (Miniforge)

Conda is required to create and manage the isolated environment (`orthogather37`) where OrthoGather will be installed. We recommend the lightweight distribution **Miniforge**, which includes Conda preconfigured with the `conda-forge` channel.

1. Check if Conda is already installed:

```
conda --version
```

If you see a version number (e.g. `conda 24.3.0`), Conda is installed and ready.

2. If Conda is not found, install Miniforge:

```
curl -L -O https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-MacOSX-arm64.sh
bash Miniforge3-MacOSX-arm64.sh
```

Follow the on-screen instructions and restart your terminal once installation completes.

3. Verify Conda installation:

```
conda --version
```

Installation procedure

1. Open a terminal in Intel (Rosetta) mode:

```
arch -x86_64 /bin/zsh
```

This ensures all Conda operations use Intel-compatible packages (`osx-64`).

2. Navigate to the OrthoGather directory (previously cloned from GitHub):

```
cd ~/OrthoGather
```

3. Grant execution permission and run the macOS installer:

```
chmod +x install_orthogather_mac.sh  
bash install_orthogather_mac.sh
```

4. What the script does automatically:

- Checks that the terminal is running in Intel/Rosetta mode.
- Detects if Conda is available — if not, it displays a clear message on how to install Miniforge.
- Validates the presence of the `environment_mac.yml` file.
- Creates or updates the Conda environment `orthogather37` using Intel packages (`CONDA_SUBDIR=osx-64`).
- Installs **OrthoFinder** and all required Python dependencies from *bioconda* and *conda-forge*.
- Verifies installation by displaying the versions of Python and OrthoFinder.

5. Once installation is complete, run OrthoGather with:

```
arch -x86_64 /bin/zsh  
conda activate orthogather37  
cd ~/OrthoGather  
python app.py
```

The local server will start at `http://127.0.0.1:5000`.

Verification

To confirm that the installation was successful:

```
python --version  
orthofinder -h
```

If both commands return version information, your environment is ready.

Summary

Rosetta + Conda (Miniforge) enables Intel compatibility on Apple Silicon. Once Conda is installed, the script automates everything — users only need to ensure that the terminal runs under Rosetta before launching OrthoGather.

5 Installation on Windows / WSL (and Linux)

This section explains how to install **OrthoGather** on Windows 10/11 using the Windows Subsystem for Linux (WSL) or directly on Linux. The process is fully automated using the script `install_orthogather_wsl.sh`, which relies on **Micromamba** for rapid and stable environment setup.

Why Micromamba is used instead of Conda

Under WSL, Conda may take a very long time (“Solving environment...”) or fail due to slow disk access and Python-based dependency resolution. **Micromamba**, a lightweight C++ reimplementation of Conda, provides the same functionality but with:

- Faster dependency solving (seconds instead of minutes)
- Smaller footprint (single binary, no base environment)
- Better performance under virtualized file systems (like WSL)

Installation procedure

1. **Open your WSL or Linux terminal and navigate to the OrthoGather folder:**

```
cd ~/OrthoGather
```

2. **Grant permission and run the installer:**

```
chmod +x install_orthogather_wsl.sh  
bash install_orthogather_wsl.sh
```

3. **First-time Micromamba installation (if required):** If Micromamba is not found, the script will display instructions:

```
curl -L https://micro.mamba.pm/install.sh | bash  
# Then close the terminal, reopen it, and run again:  
bash install_orthogather_wsl.sh
```

This step is needed only once.

4. What the script performs automatically:

- Detects WSL or native Linux environment.
- Installs Micromamba if missing.
- Creates the environment `orthogather37` with Python 3.7.
- Installs all required packages from *conda-forge* and *bioconda*.
- Validates installation by checking `python --version` and `orthofinder -h`.

5. To launch OrthoGather after installation:

```
micromamba activate orthogather37
cd ~/OrthoGather
python app.py
```

Open `http://127.0.0.1:5000` in your browser.

Verification

Confirm that the environment and OrthoFinder are functional:

```
python --version
orthofinder -h
```

Summary

Micromamba + WSL provides a reliable, fast setup for Windows and Linux users. The installer configures everything automatically; no manual environment creation is required.

6 Running OrthoGather

After the installation, **OrthoGather** can be launched directly from the terminal. All dependencies are already configured inside the environment `orthogather37` created by the automated installer.

6.1 Launching from the Terminal (Recommended)

This method is the same for macOS, Linux, and Windows via WSL.

1. Activate the environment:

```
# macOS (Intel/Rosetta)
arch -x86_64 /bin/zsh
conda activate orthogather37

# Linux or WSL
micromamba activate orthogather37
```

2. Navigate to the OrthoGather folder:

```
cd ~/OrthoGather
```

3. Start the application:

```
python app.py
```

4. Access the web interface: Open your preferred browser (Chrome, Firefox, Safari, Edge) and visit:

```
http://127.0.0.1:5000
```

Once the server starts, you should see output similar to:

```
Starting Flask on port 5000
* Serving Flask app 'app'
* Debug mode: on
* Running on http://127.0.0.1:5000/
```

6.2 Alternative: Using the Automatic Launchers

Two launcher scripts are included to simplify startup:

- `Start_Orthogather.command` — for macOS (double-click to open in Terminal)
- `start_orthogather.sh` — for Linux or WSL

Usage:

- On macOS:
 1. Open the `OrthoGather` folder.
 2. Grant execution permission (only required once):

```
chmod +x Start_Orthogather.command
```

3. Double-click the script to launch OrthoGather automatically.

- On Linux/WSL:
 1. Open a terminal and navigate to the `OrthoGather` folder:

```
cd ~/OrthoGather
```

2. Grant execution permission (only once):

```
chmod +x start_orthogather.sh
```

3. Run the launcher:

```
./start_orthogather.sh
```

These scripts:

- Automatically activate the correct environment (`orthogather37`).
- Launch the local Flask server.
- Display the access URL directly.

6.3 Optional: Creating a Desktop Shortcut

If you prefer to open the tool with a double-click:

- On macOS — drag `Start_Orthogather.command` to your Desktop.
- On Linux/WSL — right-click `start_orthogather.sh` and create a shortcut.

6.4 During Execution

Keep the terminal open while using OrthoGather. If you close the terminal, the local Flask server will stop and the browser interface will disconnect.

To stop the application safely, press:

```
CTRL + C
```

6.5 Summary

You can launch OrthoGather in three ways:

1. From the terminal (recommended for all systems)
2. Using the automatic launchers
3. Via a custom desktop shortcut

Once launched, the application will always be available at:

```
http://127.0.0.1:5000
```

Note: WSL users can access the interface directly from any Windows browser using the same address.

7 Updating OrthoGather

Keeping **OrthoGather** up to date ensures access to the latest features, bug fixes, and improvements. Updates are managed through **Git** and **Git LFS**, which allow you to synchronise your local repository with the latest version from GitHub.

7.1 When to update

You should update OrthoGather when:

- A new release or feature is announced on the GitHub repository.
- You encounter an issue that may have already been fixed.
- The installation scripts (`.sh`) or configuration files (`environment_mac.yml`, `requirements.txt`) have been modified.

7.2 Step 1: Navigate to your OrthoGather folder

Open a terminal (or WSL terminal if using Windows) and move to the folder where the tool was originally cloned:

```
cd ~/OrthoGather
```

7.3 Step 2: Pull the latest version from GitHub

Run the following commands to fetch the latest source code and large files:

```
git pull origin main
git lfs pull
```

The first command updates the repository code. The second command ensures that large binary files (e.g. `proteomes_list.json`) are updated correctly through Git LFS.

7.4 Step 3: Update dependencies (if required)

If new dependencies or package versions were introduced, the environment should be refreshed.

- **On macOS (Conda environment):**

```
arch -x86_64 /bin/zsh
conda activate orthogather37
CONDA_SUBDIR=osx-64 conda env update -f environment_mac.yml --prune
```

- **On WSL/Linux (Micromamba environment):**

```
micromamba activate orthogather37  
micromamba install -f environment.yml -y
```

or, to recreate the environment cleanly:

```
bash install_orthogather_wsl.sh
```

(This reinstalls everything automatically.)

7.5 Step 4: Verify that the update was successful

Once updated, activate the environment and run:

```
python app.py
```

If the local server launches and you can access:

```
http://127.0.0.1:5000
```

then the update was successful.

7.6 Optional: Check your environment versions

To confirm the key packages were updated:

```
python --version  
orthofinder -h  
pip list | grep flask
```

7.7 Summary of the Update Process

1. Open a terminal and go to the OrthoGather directory.
2. Run `git pull origin main` and `git lfs pull`.
3. Update the environment (via Conda or Micromamba, depending on your system).
4. Relaunch the tool to confirm everything works.

If you experience any dependency conflicts, you can always remove and recreate the environment by re-running the appropriate installation script.

For assistance with updates or troubleshooting, contact: carlos.vivasrodriguez@ucdconnect.ie

8 Conclusion

By following this guide, you now have **OrthoGather** fully installed, configured, and ready to use. All required dependencies have been automatically handled through the installation scripts — either `install_orthogather_mac.sh` (for macOS) or `install_orthogather_wsl.sh` (for Linux and Windows via WSL).

The installation process ensures that:

- A stable environment named `orthogather37` has been created with all required packages.
- All software components (*Flask*, *OrthoFinder*, *pandas*, *GOATOOLS*, etc.) are correctly installed and verified.
- Users can easily update and relaunch OrthoGather without manual dependency management.

Running and Maintaining OrthoGather

To launch the tool at any time:

```
# macOS
arch -x86_64 /bin/zsh
conda activate orthogather37
python app.py

# Linux / WSL
micromamba activate orthogather37
python app.py
```

Access the local interface at: `http://127.0.0.1:5000`

You can also use the automatic launchers:

- `Start_Orthogather.command` (macOS)
- `start_orthogather.sh` (Linux / WSL)

To keep your version up to date, simply run:

```
git pull origin main
git lfs pull
```

and, if necessary, rerun the installer script for your system.

Final Remarks

OrthoGather was designed to provide a unified, user-friendly framework for comparative proteome and gene ontology analysis. The new installation system prioritizes reproducibility and accessibility, allowing both computational biologists and non-specialist users to deploy the tool on their own machines with minimal effort.

For technical support, bug reports, or feature suggestions, please contact:
`carlos.vivasrodriguez@ucdconnect.ie`

Thank you for using **OrthoGather**. We hope this tool helps accelerate your comparative and functional proteomics research.